

Shubh Gupta

Bhopal, India

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Computer Science undergraduate at VIT Bhopal with a strong foundation in Data Science and Machine Learning. Passionate about building efficient solutions, with experience in leadership, teamwork, and event coordination.

TECH STACK

Programming: Python, C/C++, Java

Machine Learning & Data Science: PyTorch, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn

Concepts: Regression, Classification, Hyperparameter Tuning, EDA, Feature Engineering

Tools: Git, Linux/WSL, Jupyter Notebook

Web Technologies: HTML5, CSS3, JavaScript

Databases: MySQL

PROJECTS

Delhi AQI Prediction Model ([GitHub](#))

- Executed a full data science workflow, from conducting in-depth exploratory data analysis to engineering features and scaling data for model training.
- Resolved a critical overfitting issue by strategically applying hyperparameter tuning and leveraging an advanced Gradient Boosting model to ensure high-fidelity predictions.
- Built a highly accurate predictive model using XGBoost, which demonstrated a superior R2 score of 0.93+, showcasing expertise in regression modeling and ensemble techniques.
- **Utilized:** Python, Scikit-learn, Pandas, NumPy, XGBoost, Matplotlib, Seaborn

Customer Churn Prediction Model ([GitHub](#))

- Developed an end-to-end machine learning pipeline to identify at-risk customers, performing extensive EDA to uncover key churn drivers like contract type and tenure.
- Resolved significant class imbalance by implementing SMOTE and utilized a ColumnTransformer pipeline for streamlined feature scaling and encoding.
- Optimized a LightGBM classifier using GridSearchCV, achieving a strong ROC-AUC score of 0.83 and enhancing the model's ability to generalize on unseen data.
- **Utilized:** Python, Scikit-learn, LightGBM, SMOTE, Pandas, Seaborn.

AI Club landing website ([GitHub](#))

- Developed a responsive website for the AI Club at VIT University using HTML5 and CSS3, ensuring optimal user experience across various devices.
- Integrated engaging animations and interactive elements to showcase the club's mission, activities, and events, enhancing visual appeal and user engagement.
- **Utilized:** HTML5, CSS3

PUBLICATIONS

Improving Object Detection Efficiency Through Joint Information Theory and Network

Coding Techniques | SSRN Electronic Journal | Oct 2025 ([Paper](#))

- Optimized real-time object detection pipelines by integrating Random Linear and Fountain Network Coding, achieving a 9.03% accuracy improvement and 7.48% reduction in packet loss under simulated network congestion.

Real-Time Data Optimization for Object Detection Using Information Theoretic Network Coding| SSRN Electronic Journal | Dec 2025 ([Paper](#))

- Engineered a smart traffic management system using **YOLOv8** and **Sparse RLNC**, reducing intersection waiting times by **26.9%** and packet loss from **12.4% to 3.9%**.

PROFESSIONAL EXPERIENCE

Sports Club, VIT Bhopal (Football)
Vice-Captain (Team), Club Co-ordinator

Bhopal, India
April 2024–

Present

- Successfully co-ordinated major college fests including AdvITYa 2024 & 2025 and Aarambh 2024 & 2025, overseeing event planning, logistics, and execution.
- Achieved Runner-up position at AdvITYa 2025, secured 4th place at SPARDHA 2024 (IIT BHU Fest), and participated in SPORTIKON 2025 (Bennett University) and SNU Sports Fest & many more.

EDUCATION

VIT Bhopal UNIVERSITY
Bachelor of Technology in Computer Science

Bhopal, India
Sept. 2023- May 2027